1. Setting

Let $A \subseteq \mathbb{R}^{d_A}$ be the arm set and let $D \subseteq \mathbb{R}^{d_D}$ be the outcome set. We assume that A and D are Euclidean balls of radii R_A and R_D (not necessarily centred at the origin). An element $x \in A$ is called an **arm**, while an element $y \in D$ is a possible sampled outcome. The reward is affine in the arm and outcome:

$$r(x, y) := a^T x + b^T y + c.$$

Let $Z \subseteq \mathbb{R}^{d_Z}$ denote the hypothesis set, and let $z^\star \in Z$ be the unknown true hypothesis. Each hypothesis $z \in Z$ determines matrices B_z and C_z and a vector d_z of compatible dimensions. For every $z \in Z$ and $x \in A$, define

$$K_z(x) := \{y \in D : C_z y + B_z x + d_z = 0\}. \tag{1}$$

The true feasible outcome set is $K^\star(x) := K_{z^\star}(x)$.

We assume throughout that $K^\star(x)$ is nonempty for every $x \in A$.

The interaction proceeds for rounds $t = 1, ..., T$. Let $\mathcal{F}_{t-1}$ denote the history before round t. The learner chooses an arm $x_t \in A$. Nature may then choose, adaptively as a function of the history and x_t, any distribution supported on D whose conditional mean

$$m_t := E[y_t \mid \mathcal{F}_{t-1}, x_t]$$

belongs to $K^\star(x_t)$. An outcome y_t is sampled from this distribution, and the learner receives reward $r(x_t, y_t)$.

The robust value of an arm and the optimal robust value are

$$v^\star(x) := \min_{y \in K^\star(x)} r(x, y), \quad V^\star := \max_{x \in A} v^\star(x).$$

The regret over horizon T is

$$R_T := TV^\star - \sum_{t=1}^{T} r(x_t, y_t).$$

Finally, let

$$C_r := \max_{x \in A, y \in D} r(x, y) - \min_{x \in A, y \in D} r(x, y)$$

denote the reward range.

Theorem 1 (efficient $T^{\frac{8}{9}}$ learning). *Suppose that the known problem data in the setting above are rational. For every known horizon T, there is a policy whose arithmetic and weak-optimization running time is polynomial in T and the input bit length and which, for every true hypothesis and every compatible adaptive nature policy, satisfies*

$$E[R_T] \leq \tilde{O}\Big(P(d_A, d_D, R_A, R_D, \|b\|_2, C_r) T^{\frac{8}{9}}\Big),$$

where P is a fixed polynomial. No geometric property of Z is used beyond its inducing a valid family of compatible sets of the form Equation 1.

2. Warmup: Noiseless Setting

In this section, playing an arm x reveals an exact feasible response $y \in K^\star(x)$; there is no sampling noise. However, there still is the Knightian uncertainty corresponding to the adversary picking an arbitrary outcome inside $K^\star(x)$.

Lemma 1. *There exist an affine map $f : A \to \mathbb{R}^{d_D}$ and a fixed linear subspace $N \subseteq \mathbb{R}^{d_D}$ such that*

$$K^\star(x) = (f(x) + N) \cap D, \quad x \in A.$$

Equivalently, before intersecting with D, the feasible sets form a family of parallel affine spaces.

Lemma 2. *Suppose that, for every arm $x \in A$, we have identified a nonempty set $K'(x) \subseteq K^\star(x)$, and suppose that whenever the learner plays x, nature chooses an outcome in $K'(x)$. Define*

$$v'(x) := \min_{y \in K'(x)} r(x, y),$$

and choose

$$x' \in \mathrm{argmax}_{x \in A}\, v'(x).$$

Then playing x' on every round incurs nonpositive regret with respect to the original robust benchmark, i.e. $R_T \leq 0$.

Lemma 3. *There exist $d_A + 1$ arms $x^{(0)}, ..., x^{(d_A)} \in A$ such that, from any possible sequence of responses $y^{(i)} \in K^\star(x^{(i)})$, we can compute an affine map $\hat{f}$ such that*

$$K^\star(x) = \left(\hat{f}(x) + N\right) \cap D, \quad x \in A.$$

Proof. Let $x^{(0)}, ..., x^{(d_A)}$ be any affine basis of A. Play them in sequence and let $y^{(0)}, ..., y^{(d_A)}$ be the corresponding outcomes chosen by the adversary. Let $\hat{f}$ be the unique affine interpolator for which $\hat{f}(x^{(i)}) = y^{(i)}$ for all i. Since $y^{(i)} \in K^\star(x^{(i)})$, we have that $\hat{f}(x^{(i)}) + N = f(x^{(i)}) + N$ for all i. Since $x^{(i)}$'s form an affine basis, therefore $\hat{f}(x) + N = f(x) + N$ for all $x \in A$. ■

Algorithm 1: Noiseless imprecise bandit.

1. Play the $d_A + 1$ arms from Lemma 3 and use their responses to construct $\hat{f}$.
2. Initialize $N_{d_A+2} \coloneqq \{0\}$.
3. For each round $t = d_A + 2, ..., T$:
   - For every $x \in A$, define the provisional feasible set
   $$K_t'(x) \coloneqq \left(\hat{f}(x) + N_t\right) \cap D.$$
   - If $K_t'(x) = \varnothing$ for some $x \in A$, choose any such arm as x_t. Otherwise, choose
   $$x_t \in \operatorname{argmax}_{x \in A} \min_{y \in K_t'(x)} r(x, y).$$
   - Play x_t and observe y_t.
   - If $y_t - \hat{f}(x_t) \notin N_t$, set
   $$N_{t+1} \coloneqq \operatorname{span}\left(N_t \cup \left\{y_t - \hat{f}(x_t)\right\}\right).$$
   Otherwise, set $N_{t+1} \coloneqq N_t$.

Every N_t maintained by Algorithm 1 is a subspace of N. Whenever it is updated, its dimension increases by one. Moreover, if a provisional feasible set is empty, playing an arm with an empty provisional set necessarily triggers such an update. There can therefore be at most $\dim(N) \le d_D$ update rounds.

On every remaining round, all the provisional feasible sets are nonempty and the observed response belongs to $K_t'(x_t)$. Hence Lemma 2 shows that the reward on that round is at least the original robust benchmark. Charging at most a constant regret to each of the $d_A + 1$ initial rounds and to each update round gives $R_T \le O(d_A + d_D)$.

2.1. Computational Tractability

All the steps in Algorithm 1 can be carried out in polynomial time. First, affine interpolation can be done via Gaussian elimination.

The subspace N_t can be stored through an orthonormal basis. Given the new residual

$$h_t \coloneqq y_t - \hat{f}(x_t),$$

project h_t onto $N_t^\perp$. If the projection is zero, the span does not change. Otherwise, normalize the projection and append it to the stored basis. This is an incremental Gram–Schmidt and takes polynomial time.

It remains to compute the arm in the planning step. Let c_A and c_D be the centres of A and D, and let P_t be the orthogonal projector onto N_t. Define

$$q_{t(x)} \coloneqq c_D + (I - P_t)\left(\hat{f}(x) - c_D\right).$$

This is the point in $\hat{f}(x) + N_t$ closest to c_D. Consequently, $K_{t(x)}'$ is nonempty exactly when

$$\left\|q_{t(x)} - c_D\right\|_2 \le R_D,$$

and, whenever it is nonempty,

$$K_{t(x)}' = q_{t(x)} + \left\{u \in N_t : \|u\|_2 \le \sqrt{R_D^2 - \left\|q_{t(x)} - c_D\right\|_2^2}\right\}.$$

Since q_t is affine, an arm with an empty provisional fiber can be found by maximizing $\left\|q_{t(x)} - c_D\right\|_2^2$ over the ball A. This is a trust-region problem; see [1].

When every provisional fiber is nonempty, put $\beta_t \coloneqq \|P_t b\|_2$. The inner minimization over outcomes is explicit:

$$v'_{t(x)} = a^T x + b^T q_{t(x)} + c - \beta_t \sqrt{R_D^2 - \left\| q_{t(x)} - c_D \right\|_2^2}.$$

Introducing a scalar $s \geq 0$, maximizing this value over A is equivalent to the quadratic program

$$\max_{x,s} a^T x + b^T q_{t(x)} + c - \beta_t s$$

subject to

$$\|x - c_A\|_2^2 \leq R_A^2, \quad s^2 + \left\| q_{t(x)} - c_D \right\|_2^2 = R_D^2, \quad s \geq 0.$$

This problem has a fixed number of quadratic constraints. After adding the redundant ellipsoidal bound

$$\frac{\|x - c_A\|_2^2}{R_A^2} + \frac{s^2}{R_D^2} \leq 2,$$

Bienstock's fixed-constraint QCQP result [2] computes an optimizer to any desired additive accuracy in time polynomial in the input bit length and the number of requested accuracy bits. Taking accuracy T^{-2} adds at most T^{-1} to the total regret. Since the provisional model changes only when N_t is updated, the planning problem needs to be solved at most $d_D + 1$ times.

The trust-region and QCQP claims use the standard weak, additive-accuracy model. With finite-precision data, exact comparisons at the boundary of an empty fiber require the usual tolerance or gray-zone convention; this issue will be handled together with the other finite-precision errors in the noisy analysis.

3. Noisy Setting

Lemma 4. *Fix $\delta \in (0,1)$ and an integer $n \geq 1$. There exist affinely independent arms $x^{(0)}, ..., x^{(d_A)} \in A$ with the following property. Play each arm for n consecutive rounds, let*

$$|y|^{(i)} := \frac{1}{n} \sum_{s=1}^{n} y_s^{(i)}$$

be its average observed outcome, and let $\hat{f}$ be the unique affine map satisfying $\hat{f}\left(x^{(i)}\right) = |y|^{(i)}$ for every $i = 0, ..., d_A$. Then, against every compatible adaptive nature policy, with probability at least $1 - \delta$ there exists an affine map $f : A \to \mathbb{R}^{d_D}$ such that

$$K^\star(x) = (f(x) + N) \cap D, \quad x \in A,$$

and

$$\sup_{x \in A} \left\| \hat{f}(x) - f(x) \right\|_2 \leq \left(1 + 2\sqrt{d_A}\right) 2\sqrt{2} R_D \sqrt{\frac{d_D \log\left(\frac{2d_{D(d_A+1)}}{\delta}\right)}{n}}.$$

In particular, the uniform interpolation error is $\tilde{O}\left(n^{-\frac{1}{2}}\right)$.

In the noisy setting, taking the span of the observed residuals is unstable: even a small amount of noise can introduce a spurious direction. We therefore replace the subspace N_t by a learned residual ellipsoid.

Lemma 5. *Fix $\delta \in (0,1)$ and a block length n. Let $\hat{f}$ be the empirical affine interpolant constructed in Lemma 4, and put*

$$L_A := 1 + 2\sqrt{d_A}, \quad \varepsilon_{n,\delta} := 2\sqrt{2}R_D\sqrt{\frac{d_D \log\left(\frac{2d_D T}{\delta}\right)}{n}},$$
$$\eta_{n,\delta} := (1+L_A)\varepsilon_{n,\delta}, \quad R_0 := (1+L_A)R_D,$$

and

$$M_{\max} := d_D \log_2\left(1 + \frac{TR_0^2}{d_D \eta_{n,\delta}^2}\right).$$

The update rule in Algorithm 2 can be implemented in polynomial time and maintains an explicitly represented ellipsoid $\mathcal{E}_M \subseteq \mathbb{R}^{d_D}$ centred at the origin, where M is the number of blocks declared informative. Against every compatible adaptive nature policy, with probability at least $1-\delta$, there is an affine map $f : A \to \mathbb{R}^{d_D}$ satisfying

$$K^\star(x) = (f(x) + N) \cap D, \quad x \in A,$$

such that, simultaneously throughout the horizon:

- *If a block at arm x is not declared informative and $|y|$ is its empirical average outcome, then*
$$|y| - \hat{f}(x) \in \mathcal{E}_M.$$
- *The number of informative blocks satisfies $M \leq M_{\max}$.*
- *For every $\gamma \geq 0$, every $x \in A$, and every $u \in \gamma\mathcal{E}_M$,*
$$\operatorname{dist}\left(\hat{f}(x) + u, f(x) + N\right) \leq L_A \varepsilon_{n,\delta} + \gamma\left(\sqrt{M_{\max}} + 1\right)\eta_{n,\delta}.$$

In particular, the learned affine spaces have uniform error $\tilde{O}\left(n^{-\frac{1}{2}}\right)$, while only $\tilde{O}(d_D)$ blocks are informative.

Algorithm 2: Noisy ridge-ellipsoid imprecise bandit.

1. Choose the $d_A + 1$ anchor arms from Lemma 4. Play each anchor for n consecutive rounds, compute its empirical average $|y|^{(i)}$, and construct the affine interpolant $\hat{f}$. If the horizon ends during this phase, stop.
2. Set $\lambda := \rho := \eta_{n,\delta}$, with $\eta_{n,\delta}$ as defined in Lemma 5. Initialize $M := 0$ and let $\hat{G}_0$ be the matrix with no columns.
3. At the beginning of each subsequent block, define
$$V_M := \lambda^2 I + \hat{G}_M \hat{G}_M^T, \quad \mathcal{E}_M := \{u : u^T V_M^{-1} u \leq 1\}.$$
Let c_D be the centre of D, and define the expanded outcome balls
$$D_1 := \left\{y : \|y - c_D\|_2 \leq R_D + \rho\right\}, \quad D_2 := \left\{y : \|y - c_D\|_2 \leq R_D + 2\rho\right\}.$$
For every arm $x \in A$, define the inner and outer provisional fibers
$$\hat{K}_M^{\text{in}}(x) := \left(\hat{f}(x) + 2\mathcal{E}_M\right) \cap D_1,$$
$$\hat{K}_M^{\text{out}}(x) := \left(\hat{f}(x) + 3\mathcal{E}_M\right) \cap D_2.$$
4. If $\hat{K}_M^{\text{in}}(x) = \varnothing$ for some $x \in A$, choose any such arm. Otherwise, choose
$$x \in \operatorname{argmax}_{x' \in A} \min_{y \in \hat{K}_M^{\text{out}}(x')} r(x', y).$$
5. Play the chosen arm for n rounds. If fewer than n rounds remain, play until the horizon and stop without updating. Otherwise, let $|y|$ be the block-average outcome and set
$$\hat{g} := |y| - \hat{f}(x).$$
6. If
$$\hat{g}^T V_M^{-1} \hat{g} > 1,$$
declare the block informative, set $\hat{G}_{M+1} := \left[\hat{G}_M, \hat{g}\right]$, and then increase M by one. Otherwise leave $\hat{G}_M$ and M unchanged. Return to Step 3 and continue until time T.

References

[1] J. J. Moré and D. C. Sorensen, “Computing a Trust Region Step,” *SIAM Journal on Scientific and Statistical Computing*, vol. 4, no. 3, pp. 553–572, 1983, doi: 10.1137/0904038.

[2] D. Bienstock, “A Note on Polynomial Solvability of the CDT Problem,” *SIAM Journal on Optimization*, vol. 26, no. 1, pp. 488–498, 2016, doi: 10.1137/15M1009871.